

Week 4 Initial Results: CVaR-MPPI for Highway Lane-Change Planning

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I. MAIN IDEA

A risk-sensitive lane-change planner that augments Model Predictive Path Integral (MPPI) with a Conditional Value-at-Risk (CVaR) objective. Augmentation of CVaR replaces the risk-neutral expectation over rollouts with a sampled tail-risk backup. The resulting controller suppresses lane-changes that are efficient on average but unsafe under rare traffic outcomes.

II. RELATED WORK

A. Overview

In one paragraph, describe the relevant literature that you have reviewed or are planning to review for your project. If you have not yet identified sufficient literature, describe several categories of literature that you intend to explore and explain why they are relevant. *You can copy this from your Week 2 report if not revised (not graded).*

B. Highlighted Literature Review

In this section, provide a more detailed summary of a single paper that is most relevant to your project direction. *You can copy this from your Week 2 report if not revised (not graded).*

III. PROBLEM FORMULATION

Let $x_k \in \mathbb{R}^{n_x}$ be the joint ego-traffic state, $u_k = v_k + \epsilon_k \in \mathbb{R}^{n_u}$ the perturbed ego control with $\epsilon_k \sim \mathcal{N}(0, \Sigma_\epsilon)$, and w_k exogenous traffic noise. The simulated stochastic dynamics $\tilde{x}_{k+1} = \tilde{F}(\tilde{x}_k, u_k, w_k)$ form an unbiased estimate of the nominal $x_{k+1} = F(x_k, u_k)$ [5]. Over a horizon K with state cost q , terminal cost ϕ , and risk cost $L(\tilde{x}) = \sum_{k=0}^{K-1} \ell(\tilde{x}_k)$, we seek

$$\min_v \mathbb{E} \left[\phi(x_K) + \sum_{k=0}^{K-1} \left(q(x_k) + \frac{\lambda}{2} v_k^\top \Sigma_\epsilon^{-1} v_k \right) \right] \quad (1)$$

s.t. $\text{CVaR}_\alpha(L(\tilde{x})) \leq C_u$,

where $\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]$ is the conditional expectation of the worst $1 - \alpha$ tail [2], $\alpha \in (0, 1)$ the confidence level, $\lambda > 0$ the MPPI temperature, and C_u caps the admissible tail risk.

IV. SKETCH OF METHOD

A. Offline Component

The offline stage instantiates the dynamic-programming foundation of CVaR control by Chow et al. [4]. Augmenting the state with a confidence variable $y \in (0, 1]$, the CVaR-MDP value $V^*(x, y) = \min_{\mu \in \Pi_H} \text{CVaR}_y(\sum_t \gamma^t Z_t \mid x_0 = x, \mu)$ admits a γ -contractive, concavity-preserving Bellman recursion

$$\mathbf{T}[V](x, y) = \min_a [c(x, a) + \gamma \max_{\xi \in \mathcal{U}_{\text{CVaR}}(y, P)} \sum_{x'} \xi(x') V(x', y \xi(x')) P(x' \mid x, a)] \quad (2)$$

[4]. We discretize a coarse three-lane MDP over (lane index, gap-to-lead, ego speed) on $\sim 10^3$ cells with 21 log-spaced confidence points and run interpolated value iteration (Algorithm 1 of [4]) to obtain $V^*(x, y)$, which is loaded as the MPPI terminal cost $\phi(x_K) \leftarrow V^*(\Pi_{x_K}, y)$, where Π_{x_K} maps the continuous terminal state to the coarse MDP grid. This provide a tail-risk-shaped cost-to-go beyond the rollout horizon. Offline complexity is $\mathcal{O}(|\mathcal{X}| |\mathcal{Y}| |\mathcal{A}| C_\zeta n_{\text{iter}})$ where C_ζ is the cost of the inner risk-envelope optimization. This computational cost is paid once before deployment; runtime queries are linear-interpolation lookups in microseconds.

B. Online Component

The online stage executes a Risk-Aware MPPI receding-horizon controller [1], [5]. At each step we sample M control sequences $u_k^m = v_k + \epsilon_k^m$, propagate K -step rollouts under the nominal bicycle to accumulate S_m , and for each m simulate N stochastic traffic futures via $\tilde{F}$ to compute the Monte-Carlo CVaR estimate

$$\widehat{\text{CVaR}}_\beta^m = \frac{1}{N_o} \sum_{n=0}^{N-1} \mathbf{1}_{\{L(\tilde{x}^{m,n}) \geq \widehat{\text{VaR}}_\beta^m\}} L(\tilde{x}^{m,n}), \beta \in (0, 1) \quad (3)$$

empirical tail-average CVaR estimator used in RA-MPPI. [5]. The penalty is folded into a modified rollout cost $\tilde{S}_m = \widehat{\text{ACVaR}}_\beta^m \mathbf{1}_{\{\widehat{\text{CVaR}}_\beta^m > C_u\}} + S_m$, importance weights $\tilde{\omega}_m = \exp(-(\tilde{S}_m - \min_m \tilde{S}_m)/\lambda)$ are formed, and the mean control is updated as $v^+ = \sum_m \tilde{\omega}_m u^m / \sum_m \tilde{\omega}_m$; the first input v_0^+ is applied and $v \leftarrow v^+$ warm-starts the next step. This stage is suitable because GPU-parallel rollouts handle the continuous nonlinear bicycle within the ~ 100 ms control budget, and Monte-Carlo CVaR accommodates arbitrary, non-Gaussian traffic disturbances. It complements the offline stage by

injecting fine-grained interaction with current surrounding vehicles that the coarse V^* cannot resolve, while V^* supplies the long-horizon CVaR-Bellman backup that a finite- K rollout would otherwise miss.

V. INITIAL RESULTS

A. Modeling & Simulation

Describe the *model* of the running example system on which you will deploy your method. *Please start off from what you wrote in Week 3 report.*

B. Baseline methods

Describe the methods against which you are comparing.

C. Initial Results

Report and explain your initial results in *one paragraph*.

VI. NEXT STEPS

Describe what steps you are planning next to improve your results and develop your research direction further.

VII. QUESTIONS

If you have any questions that you would like feedback on from your classmates or the instructor, please share them here.

REFERENCES

- [1] G. Williams, A. Aldrich, E. A. Theodorou, "Model predictive path integral control: From theory to parallel computation," *J. Guid. Control Dyn.*, vol. 40, no. 2, pp. 344–357, 2017.
- [2] R. T. Rockafellar, S. Uryasev, "Optimization of conditional value-at-risk," *J. Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [3] Y. Chow, M. Ghavamzadeh, "Algorithms for CVaR optimization in MDPs," in *NeurIPS*, vol. 27, 2014.
- [4] Y. Chow, A. Tamar, S. Mannor, M. Pavone, "Risk-sensitive and robust decision-making: a CVaR optimization approach," in *NeurIPS*, vol. 28, 2015.
- [5] J. Yin, Z. Zhang, P. Tsiotras, "Risk-aware model predictive path integral control using conditional value-at-risk," arXiv:2209.12842, 2022.